

1. Abstract

The photoelectric effect was investigated using a phototube and a set of 8 LEDs to determine Planck constant, work function of metal, and intensity dependencies. By modeling stopping voltage as a function of emission frequency between 450 THz to 760 THz, the experiment yielded an experimental Planck constant of $h = (4.623 \pm 0.005) \times 10^{-34}$ Js, a threshold voltage of $f_0 = 304.0 \pm 0.7$ THz, and work function $E_0 = (1.405 \pm 0.007) \times 10^{-19}$ J.

While the data followed the linear trend predicted by Einstein's photoelectric equation

$V_{stop} = (h/e)(f - f_0)$, the experimental value of h is 401 error bars below the actual Planck constant.

The deviation is highlighted by a high reduced χ^2 of 7200.9. This value means the input uncertainties were severely underestimated.

Furthermore, changing the intensity of a single LED confirmed the photocurrent and intensity of light are directly proportional. However, the stopping voltage was not independent of intensity. Instead, it showed a decreasing exponential trend. The deviation is likely due to physical factors such as LED thermal spectral shifting and space charge effects at higher output levels.

2. Introduction (Sources: PHY224 Lab Manual, National Institute of Standards and Technology):

In 1899, J.J. Thomson discovered the photoelectric effect: illuminating light on metal causes the metal to launch negative charges. This experiment investigates such an effect by using an apparatus with phototubes and 8 LEDs as light sources. The objective is measure the stopping voltage required to halt photocurrent across various incident light frequencies. By analyzing this relationship, the experiment aims to determine Planck constant h , work function of metal E_0 , the threshold frequency f_0 of metal, and evaluate the dependence of both stopping voltage and photocurrent on light intensity.

Initial explanations on the photoelectric effect relies on Maxwell's wave theory. This theory predicted that if the metal was irradiated by a very dim light, there will be a delay for electrons to absorb enough energy to overcome metal's work function E_0 . However, studies by Philip Leanard observed that photocurrent appeared instantaneously regardless of the light's dimness. Furthermore, he found that electrons were only emitted if the incident light passed a specific threshold frequency f_0 and the maximum kinetic energy of the emitted electron depended only on the light's frequency, not its intensity.

In 1905 to solve the contradiction, Einstein proposed that the energy of light is not continuous, but quantized into discrete packets named photons. The energy in a single photon (E) is directly proportional to its frequency (f): $E = hf$, where h is Planck constant (6.626×10^{-34} Js). When a photon hits the metal, the energy is transferred to an electron. If such energy is higher than the metal's work function (E_0), the electron will escape the bond and the excess energy becomes the kinetic energy of the escaped electron (E_k). Substitute Einstein's photon energy into the classical kinetic equation gives $E_k = hf - E_0$. To

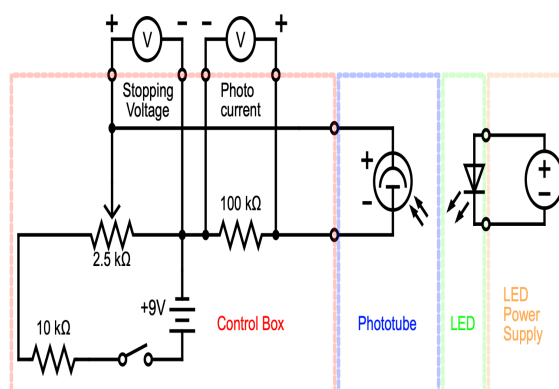
experimentally measure the kinetic energy, a stopping voltage V_{stop} is applied to the detector to repel the launched electron: $V_{stop} = E_k/e$, where e is the elementary charge (1.602×10^{-19} C). Further derivation gives $V_{stop} = (h/e)(f - f_0)$.

3. Methods

Hand-drawn circuit diagram for a photoconductor experiment. The diagram includes the following components and connections:

- Power Supply:** A central box labeled "power supply" with four output terminals.
- Multimeter 1:** Labeled "Multimeter 1", it has two input terminals. One terminal is connected to the positive output of the power supply, and the other is connected to the negative output of the power supply.
- Multimeter 2:** Labeled "Multimeter 2", it has two input terminals. One terminal is connected to the positive output of the power supply, and the other is connected to the negative output of the power supply.
- Phototube:** A box labeled "phototube" with two terminals. One terminal is connected to the positive output of the power supply, and the other is connected to the negative output of the power supply.
- LED driver circuit:** A box labeled "LED driver circuit" with two terminals. One terminal is connected to the positive output of the power supply, and the other is connected to the negative output of the power supply.
- Photoconductor:** A box labeled "photoconductor" with two terminals. One terminal is connected to the positive output of the power supply, and the other is connected to the negative output of the power supply.

Below the photoconductor box, the text $(\text{voltage} / \text{h} \nu) - \text{threshold} = 0$ is written.



In Exercise 1, each LEDs was operated in CW mode with intensity set to a fixed level (controlled by a knob on the driver). Starting with stopping voltage set to 0 or all the way counter clockwise, slowly turn the knob on the control box clockwise. Stop when the photocurrent reading starts jumping around 0. Record the reading on the stopping voltage multimeter. The stopping voltage measurement was repeated 5 times for each LEDs in order to estimate the spread of the value. The mean of the five trials was reported as V_{stop} , and the standard deviation was taken as the uncertainty.

The LED frequencies can be obtained from their central wavelength using $f = c/\lambda$ (wave length uncertainties were provided by the manufacturers). The data set $\{f, V_{stop}\}$ was then fitted with into linear model $V_{stop} = h/e(f - f_0)$ to determine the h, E_o , and f_0 .

In Exercise 2, a single LED (wavelength = 455nm) was selected, and the circuit set up remained unchanged. The LED intensity was varied with multiple levels using the LED driver (CW mode, divided the intensity level knob on the driver into six levels with equal gap in between. For each intensity setting, the photocurrent was recorded at $V_{stop} \approx 0$ V. The stopping voltage was then measured by reducing the photocurrent to zero (note: that all the procedures for determining zero are the same as Exercise 1). These procedures were repeated for all intensity levels. The photocurrent was calculated as $I = V_R / 100 \text{ k}\Omega$ as needed. After that, the data $\{I, V_R\}$ was collected and plotted, suitable fitting model was used to find out whether the photocurrent is proportional to the LED intensity and whether the V_{stop} was approximately unchanged with the change of intensity.

4. Results and Plots

When placing LEDs with 940 nm into the phototube, the multimeter reading for photocurrent stays as 0. Thus, the data point is removed (more in discussion).

frequency (THz)	Vstop (V)	$\Delta(V_{stop})$ (V)
760	1.2796	0.0006
660	1.1698	0.001
600	0.859	0.001
570	0.817	0.004
510	0.630	0.007
486	0.501	0.006
450	0.423	0.004

Table 1: Exercise 1 Data (Frequency vs. Stopping Voltage) Experimental Data for the photoelectric effect showing the relationship with incident light frequency and its corresponding stopping voltage required to bring photocurrent to 0. Measurements were taken using 7 distinct LEDs, ranging from 450Thz to 760Thz.

Intensity (% of LED driver knob)	Photocurrent Calculated (μA)	$\Delta(\text{Photocurrent Calculated})$ (μA)	Vstop (V)	$\Delta(V_{stop})$ (V)
16.7	0.2004	0.0003	1.26720	0.00003
33.3	0.8000	0.0003	1.17000	0.00003
50	1.16000	0.00009	1.15945	0.00003
66.7	2.2400	0.0001	1.13708	0.00003
83.3	3.0500	0.0001	1.11775	0.00003
100	3.2900	0.0002	1.11103	0.00003

Table 2: Exercise 2 Data (Intensity vs. Photocurrent & Stopping Voltage) Measurement of photocurrent and stopping voltage across varying intensity for a single light source of wavelength = 455nm. Intensity was controlled by rotating the knob on the LED driver, represented as a percentage of the driver's maximum output.

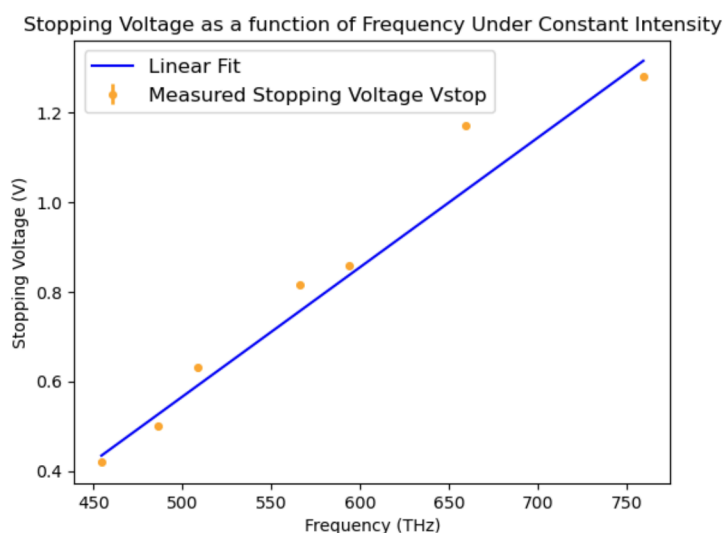


Figure 1: Stopping voltage as a function of incident light frequency.

(Table 1)The data is fitted to a linear model (blue) based on Einstein's photoelectric equation,

$V_{stop} = h/e(f - f_0)$. The fit yielded experimental

$$h = (4.623 \pm 0.005) \times 10^{-34} \text{ Js,}$$

$$f_0 = 304.0 \pm 0.7 \text{ THz, and}$$

$$E_0 = (1.405 \pm 0.007) \times 10^{-19} \text{ J.}$$

Figure 2: Residuals of the linear fit for stopping voltage vs frequency (Figure 1).

The random distribution around the baseline indicates the linear fit is consistent. However, the large reduced χ^2 of 7200.9 yielded by the data shows the estimated uncertainty from the multimeter specification sheet itself is too small (more in discussion).

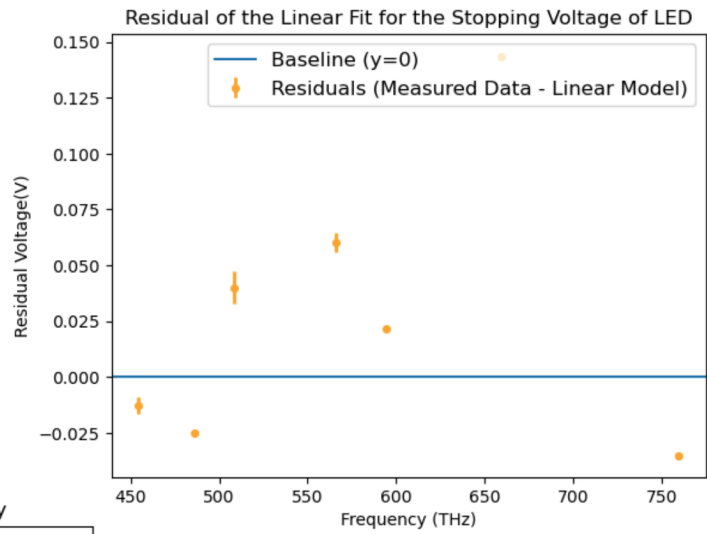


Figure 3: Stopping voltage as a function of relative LED intensity. (Table 2) Different from ideal theoretical models which predict a constant stopping voltage, the measured stopping voltage shows a decreasing trend. To highlight this deviation, the data is fitted with an exponential model and yielded : $V_{stop} = (0.03177 \pm 0.0002 \text{ V}) \times (0.95598 \pm 0.00005)^{\text{intensity}} + (1.11350 \pm 0.00003 \text{ V})$

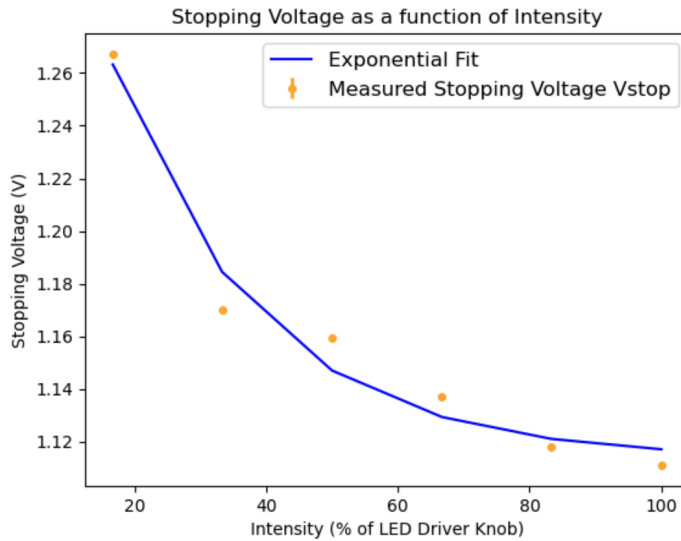
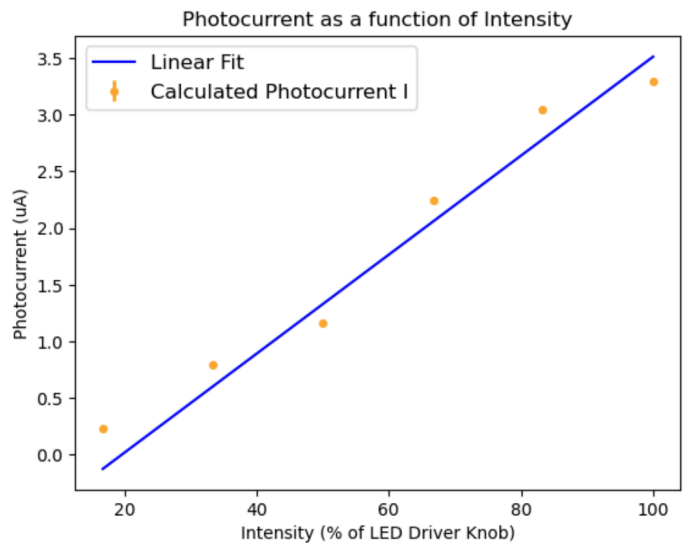


Figure 4: Photocurrent as a function of relative LED intensity. (Table 2) The data is fitted with an linear model and yielded: $I = ((8.527 \pm 0.002) \times 10^{-8} \text{ A/\% knob}) \times \text{intensity} - (8.527 \pm 0.002) \times 10^{-7} \text{ A}$

The data confirms the expectation that photocurrent is directly proportional to the intensity of incident light.



5. Uncertainties

When measuring the stopping voltage, the precision of multimeter reading is given by the specification sheet(Appendix). Furthermore, photocurrent did not decrease linearly as the applied voltage approached stopping voltage. As a result, there was a spread in the value of V_{stop} . Thus, the uncertainty of V_{stop} is calculated by taking the standard deviation of the 5 trials.

The theoretical model assumes a perfectly monochromatic light source. However, LEDs emit a range of frequency around a central peak, introducing a systematic error. Each LED's range of wavelength is provided by the manufacturers (can also be found by looking at the FWHM)(Appendix)..

When converting wavelength to frequency, uncertainty propagation is used: $\Delta f = c/\lambda^2 \times \Delta \lambda$.

When converting photocurrent readings in voltage to current, uncertainty propagation is used:

$$\Delta(I) = \sqrt{(\Delta(V)/V)^2 + (\Delta(R)/R)^2}.$$

Space charge effects at high intensity and contact potential difference are explained in the discussion section.

6. Analysis & Discussion

After fitting frequency of LEDs and their corresponding stopping voltage (Table 1) into Einstein's model:

$V_{stop} = h/e(f - f_0)$, the experimental value of $h = (4.623 \pm 0.005) \times 10^{-34}$ Js, threshold voltage $f_0 = 304.0 \pm 0.7$ THz, and work function $E_0 = (1.405 \pm 0.007) \times 10^{-19}$ J.

While the random distribution of residuals around baseline in Figure 2 shows the relationship between frequency and stopping voltage does a linear trend, the yielded h is 401 error bars below the actual Planck constant 6.626×10^{-34} Js. The large deviation in h is supported by the large reduced χ^2 of 7200.9. This means while the model accurately describes the relationship between frequency and stopping voltage, the experiment underestimated the uncertainty in the set up.

Specifically, the cathode and anode within the phototube (metal and detector) are often built with different materials, which possess different work functions. This creates a small potential difference between the 2 metals even when no stopping voltage is applied. Such an effect systematically underestimated V_{stop} .

This is also why when no stopping voltage is applied, there is still approximately 0.003V in the multimeter reading. Since V_{stop} and h are directly proportional in the model, experimental h is also underestimated. This explains why the experimental h is 401 error bars below actual value.

When the 940 nm LED is placed in the phototube, the reading of photocurrent stays at 0, meaning no electron is ejected from the metal. Which is reasonable because the LED's frequency of 319 THz is too close to the experimental $f_0 = 304.0 \pm 0.7$ THz. Since the LED is not useful to determine the stopping voltage (photocurrent is already at 0), the data point is removed.

Exercise 2 isolated a single LED to test how light intensity affects photocurrent and stopping voltage.

When plotting the intensity vs. photocurrent out, a linearly increasing trend is shown (Figure 4). When the data is fitted to basic linear model, it yielded the following parameters: $I = ((8.527 \pm 0.002) \times 10^{-8} \text{ A/\% knob}) \times \text{intensity} - (8.527 \pm 0.002) \times 10^{-7} \text{ A}$. Such a directly proportional relationship is consistent with the experiment's prediction.

On the other hand, when plotted out intensity vs stopping voltage (Figure 3), an exponentially decreasing trend is shown, and yielded parameters $V_{stop} = (0.03177 \pm 0.0002 \text{ V}) \times (0.95598 \pm 0.00005)^{\text{intensity}} + (1.11350 \pm 0.00003 \text{ V})$ when fitted to the basic exponential model. This is inconsistent with the experiment's prediction of independent relationship, meaning there must be some secondary physical effects. In fact, when increasing the intensity of the LED, the metal ejects a dense cloud of electrons. These negatively charged cloud naturally repels newly ejected electrons, which provide "free" stopping power and results in lower V_{stop} from the potentiometer. This explains why the stopping voltage decreases as intensity increases.

7. Conclusion

This experiment investigated the photoelectric effect using a vacuum phototube and eight LEDs with frequencies ranging from 450 THz to 760 THz. In Exercise 1, stopping voltages were measured for multiple LED frequencies and analysed with the Einstein's photoelectric model: $V_{stop} = h/e(f - f_0)$. The data confirmed the linear relationship between frequency and stopping voltage, the model yielded $h = (4.623 \pm 0.005) \times 10^{-34} \text{ Js}$, $f_0 = 304.0 \pm 0.7 \text{ THz}$, and $E_0 = (1.405 \pm 0.007) \times 10^{-19} \text{ J}$. The underestimation of experimental h (401 error bars below actual value and large reduced χ^2 of 7200.9 implies some uncertainties are underestimated such as the contact potential differences.

In Exercise 2, the intensity of a single LED ($\lambda = 455 \text{ nm}$) was varied. As predicted, the photocurrent increased with the increasing intensity. However, the measured stopping voltage showed a small decrease with the increasing intensity rather than the experiment's prediction of remaining constant. Such deviation from prediction reflects non-ideal factors, such as the space charge effect.

Overall, the experiment qualitatively supports Einstein's photocurrent model. Some limitations were identifying the zero-line of the V_{stop} (The range of fluctuations around 0) which generates objective uncertainties, and systematic errors caused by secondary physical effects, which is hard to capture.

Appendix:

Uncertainty of Multimeter Reading based on specification sheet:

For a range of 100mV, the uncertainty is $(0.003 + 0.003)$, which means 0.003% of reading + 0.003% of the range (100mV). For the range of 1V, the uncertainty is $(0.002 + 0.0006)$, which means 0.002% of reading + 0.0006% of the range (1V)

Uncertainty of LEDs wavelength based on manufacturers:

LED wavelength (nm)	u(wavelength) (nm)
395	15
455	16
505	30
530	35
590	15
617	12
660	14

Model Number of LEDs:

Wavelength of LED	395nm	455nm	505nm
Model #	M01265872	M01233423	M01252531

530nm	590nm	617nm	660nm	940nm
M01241344	M01228449	M01256542	M01255278	M01254404